

PRAKRITI SARKAR

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Computer Science Engineering Student · Software Development · Machine Learning · Web Development

SUMMARY

Computer Science Engineering student at Sister Nivedita University with hands-on experience building machine learning systems, full-stack web applications, and blockchain-based interfaces. Demonstrated ability to design, develop, and deploy end-to-end solutions that address real-world problems. Strong foundation in Python, JavaScript, and data-driven development with experience collaborating on technical projects and delivering functional deployed solutions.

SKILLS

Languages: Python, C, JavaScript

Web Technologies: HTML, CSS, Tailwind CSS, Next.js, Streamlit

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn, Joblib, Ethers.js, Web3.js

Databases: MongoDB, Oracle Database

Tools & Platforms: Git, GitHub, MetaMask, Solidity (Ethereum smart contracts)

Other: Technical Writing, Editing, Agile Problem Solving, Design Thinking, Communication, Collaboration

PROJECTS

Wellness Personas of SNU

2024-01 – 2024-06

Python, Streamlit, KMeans Clustering, Scikit-learn, Pandas, Matplotlib

- Developed a machine learning pipeline using KMeans clustering to segment users into distinct lifestyle personas, enabling structured analysis of behavioral patterns
- Built and deployed an interactive Streamlit interface that processes multi-dimensional user data and generates visualizations for intuitive exploration of clustering results
- Automated persona generation from raw survey data, reducing manual analysis effort and supporting insights for the SNU user base

BlockVault — Decentralized Lending Prototype

2024-07 – 2024-10

Next.js, Tailwind CSS, Solidity, Ethers.js / Web3.js, MetaMask

- Designed and implemented a decentralized lending prototype that simulates on-chain credit profiles derived from wallet borrowing and repayment history, addressing limitations of collateral-only DeFi lending
- Integrated Web3 wallet connectivity using MetaMask, enabling users to connect wallets, submit loan requests, and interact with Ethereum smart contracts through a responsive Next.js frontend
- Deployed smart contracts and validated end-to-end loan lifecycle transactions within a demo environment

SplitX — Smart Group Expense Tracker

2024-11 – 2025-02

HTML, CSS, JavaScript, Tailwind CSS, Python, MongoDB

- Built and deployed a full-stack web application that records group expenses, automatically calculates member balances, and generates settlement summaries, eliminating manual calculation errors
- Implemented a Python backend with MongoDB integration to store transaction records and compute dynamic net balances across multi-member groups
- Designed a responsive Tailwind CSS frontend delivering real-time expense updates and a structured settlement dashboard

EDUCATION

Sister Nivedita University

2023-07 – 2027-07

Bachelor of Technology (B.Tech), Computer Science Engineering

CGPA: 8.36

CERTIFICATIONS

- Algorithms for Searching, Sorting, and Indexing
- Generative AI: Introduction and Applications

CLUBS & ACTIVITIES

LNC Tech Community

2025-01 – Present

Creation Team Lead

- Lead a team of four content writers producing blog posts and technical community content
- Coordinate content planning, collaboration, and publishing to maintain consistent engagement within the tech community

Aperture Alchemist Photography Club

2024-01 – 2025-01

Content Writer

- Produced captions, blog posts, and promotional content for photography events and club activities
- Developed social media content and supported promotion of multiple photography events organized within the college